DTC DETECTION LOGIC AND CONDITIONS [SKYACTIV-D 1.5]

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DTC No.	Condition	Detection condition
B10A2:00	Vehicle collision	• A collision signal from the SAS control module is received.
P0016:00	Camshaft position/Crankshaft position correlation problem	• CKP sensor and CMP sensor input signal does not match while the crankshaft rotates **15 times** when the following condition A or B is met: **Condition A:** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — Engine speed: **1,000 rpm or less** — Engine coolant temperature: **-10 °C {14 °F} or more** — The following DTCs are not detected: • CKP sensor: P0336:00, P0337:00 • CMP sensor: P0341:00, P0342:00 **Condition B:** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — Engine speed: **700—1,000 rpm or less** — Engine coolant temperature: **-10 °C {14 °F} or more** (engine on) — The following DTCs are not detected: • ECT sensor: P0117:00, P0118:00 • CKP sensor: P0336:00, P0337:00 • CMP sensor: P0341:00, P0342:00
P0030:00	A/F sensor heater control circuit range/performance problem	• With the following conditions met, a condition is detected by the PCM in which the voltage while the A/F sensor heater is operating or not operating is out of the specified value for **5 s**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — A/F sensor heater control duty value: **10—90 %**
P0047:00	Regulating solenoid valve control circuit low input	• If the PCM detects that the turbocharger solenoid valve voltage at the PCM terminal 1CH is less than the specified value for **1 s** with the following condition met, the PCM determines that the turbocharger solenoid valve circuit voltage is low. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0048:00	Regulating solenoid valve control circuit high input	• If the PCM detects that the turbocharger solenoid valve current at the PCM terminal 1CH is **3.5 A or more** for **1 s** with the following condition met, the PCM determines that the turbocharger solenoid valve circuit voltage is high. **MONITORING CONDITIONS** — Battery voltage: **8—16 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0072:00	Ambient temperature sensor circuit low input	• The PCM monitors the input signal from the ambient temperature sensor. If the voltage from the ambient temperature sensor is **below 0.13 V** for **5 s**, the PCM determines that the ambient temperature sensor circuit has a malfunction.
P0073:00	Ambient temperature sensor circuit high input	• The PCM monitors the input signal from the ambient temperature sensor. If the voltage from the ambient temperature sensor is **above 4.9 V** for **5 s**, the PCM determines that the ambient temperature sensor circuit has a malfunction.

DTC No.	Condition	Detection condition
P007B:00	Boost air temperature sensor circuit range/performance problem	• With the following conditions met, a condition has continued for **15 s** in which difference in intake air temperature measured by the IAT sensor No. 1 and boost air temperature sensor is **150 °C {270 °F} or more** and difference in intake air temperature measured by boost air temperature sensor and IAT sensor No.2 is **150 °C {270 °F} or more**. **MONITORING CONDITIONS** — Period vehicle being left: **6 h or more** — Battery voltage: **10 V or more** — **Within 20 s** after engine start — The following DTCs are not detected: • Boost air temperature sensor: P007C:00, P007D:00 • IAT sensor No.1: P0112:00, P0113:00
P007C:00	Boost air temperature sensor circuit low input	• If the PCM detects that the boost air temperature sensor voltage at the PCM terminal 1N is **0.15 V or less** for **5 s** with the following condition met, the PCM determines that the boost air temperature sensor circuit voltage is low. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P007D:00	Boost air temperature sensor circuit high input	• The PCM monitors the input signal from the boost air temperature sensor. If the voltage from the boost air temperature sensor is **above 4.94 V** for **5 s**, the PCM determines that the boost air temperature sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0087:00	Low pressure malfunction in common rail fuel pressure control system	• With the following conditions met, a condition has continued for **15 s** in which the fuel pressure sensor output value is lower than the fuel pressure control target value by **20 MPa {204 kgf/cm^2, 2901 psi}**: — The following conditions continue for **2 s**: • Battery voltage: **8 V or more** • Fuel pressure control: Being performed • Change in target fuel pressure during half-rotation of crankshaft: **-6 to 6 MPa {-61—61 kgf/cm^2, -870—870 psi}** • Fuel temperature: **-25—70 °C {-13—158 °F}** • Target fuel discharge amount from supply pump: **Less than 10,000 mm^3/rev** — The following DTCs are not detected: • Fuel temperature sensor: P0181:00, P0182:00, P0183:00 • Fuel pressure sensor: P0191:00, P0192:00, P0193:00 • Suction control valve: P062A:00
P0088:00	High pressure malfunction in common rail fuel pressure control system	• With the following conditions met, a condition has continued for **15 s** in which the fuel pressure sensor output value is higher than the fuel pressure control target value by **20 MPa {204 kgf/cm^2, 2901 psi}**: — The following conditions continue for **2 s**: • Battery voltage: **8 V or more** • Fuel pressure control: Being performed • Change in target fuel pressure during half-rotation of crankshaft: **-6 to 6 MPa {-61—61 kgf/cm^2, -870—870 psi}** • Fuel temperature: **-25—75 °C {-13—167 °F}** • Target fuel discharge amount from supply pump: **Less than 10,000 mm^3/rev** — The following DTCs are not detected: • Fuel temperature sensor: P0181:00, P0182:00, P0183:00 • Fuel pressure sensor: P0191:00, P0192:00, P0193:00 • Suction control valve: P062A:00

DTC No.	Condition	Detection condition
P0089:00	High pressure malfunction in common rail fuel pressure control system	• When the following conditions are met, the actual fuel pressure **exceeds 217 Mpa {2,213 kgf/cm^2, 31,473 psi}** for a continuous **3 s**: **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — The following DTCs are not detected: • Fuel pressure sensor: P0191:00, P0192:00, P0193:00
P0093:00	Fuel leakage from fuel pressure control system	• When the following conditions are met, the PCM detects the amount of fuel leakage as **exceeding 480 mm^3/rev 3 times**: **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — Engine speed: **600 rpm or more** — Fuel-cut control is not implemented — Fuel pressure relief valve: Closed — Fuel pressure control: Being performed — CKP sensor: Normal — The following DTCs are not detected: • Fuel pressure sensor: P0191:00, P0192:00, P0193:00 • Suction control valve: P062A:00 • Fuel injector: P0201:00, P0202:00, P0203:00, P0204:00, P1378:00, P1379:00, P2146:00, P2147:00, P2148:00, P2149:00, P2150:00, P2151:00 • Common rail fuel pressure control system: P0089:00 • Fuel injector code not programmed: P1675:00, P1676:00 • PCM internal malfunction: P062B:00
P0096:00	IAT sensor No.2 circuit range/ performance problem	• With the following conditions met, a condition has continued for 15 s in which difference in intake air temperature measured by the boost air temperature sensor and IAT sensor No.2 is **150 °C {270 °F} or more** and difference in intake air temperature measured by IAT sensor No.1 and IAT sensor No.2 is **150 °C {270 °F} or more**. **MONITORING CONDITIONS** — Period vehicle being left: **6 h or more** — Battery voltage: **10 V or more** — Within **20 s** after engine start — The following DTCs are not detected: • Boost air temperature sensor: P007C:00, P007D:00 • IAT sensor No.1: P0112:00, P0113:00 • IAT sensor No.2: P0097:00, P0098:00
P0097:00	IAT sensor No.2 circuit low input	• If the PCM detects that the IAT sensor No.2 voltage at the PCM terminal 1CQ is **0.05 V or less** for **5 s** with the following condition met, the PCM determines that the IAT sensor No.2 circuit voltage is low. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0098:00	IAT sensor No.2 circuit high input	• The PCM monitors the input signal from the IAT sensor No.2. If the voltage from the IAT sensor No.2 is **above 4.93 V** for **5 s**, the PCM determines that the IAT sensor No.2 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)

DTC No.	Condition	Detection condition
P009B:00	Fuel pressure relief valve control circuit open	• When the following condition is met, the PCM detects the control current at fuel pressure relief valve as **0.67 A or less** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — **0.42 msec or more** have elapsed since fuel pressure relief pressure operation (fuel pressure reduction) — **2 s or more** have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P009C:00	Fuel pressure relief valve control circuit low input	• When the following condition is met, control voltage at fuel pressure relief valve as **1.2 V or less** for countinuous **1 s** (engine off) or continuous **8 times** (engine on). **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **2 s or more** have elapsed since PCM launch — Any one of the following conditions is met: • Fuel pressure when engine speed is **1,800 rpm or more** is **211 MPa {2152 kgf/cm^2, 30604 psi} or more** • Engine speed of **3,300 rpm or less** (also when vehicle is stopped) — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P009D:00	Fuel pressure relief valve control circuit high input	• When the following condition is met, control voltage at fuel pressure relief valve as 4.8 V or more for countinuous 1 s (engine off) or continuous 8 times (engine on). **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **2 s** or more have elapsed since PCM launch — Any one of the following conditions is met: • Fuel pressure when engine speed is **1,800 rpm or more** is **211 MPa {2152 kgf/cm^2, 30604 psi} or more** • Engine speed of **3,300 rpm or less** (also when vehicle is stopped) — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P009E:00	Fuel pressure relief valve malfunction (stuck close)	• When the following conditions are met, the pressure reduction amount during the fuel pressure relief valve operation is insufficient for several seconds continuously. **MONITORING CONDITIONS** — Battery voltage: **10—16 V** — **0.5 s** have elapsed after ignition was switched ON. — During fuel pressure control — The following DTCs are not detected: • Fuel pressure control system: P0093:00 • Fuel pressure relief valve control: P009B:00, P009C:00, P009D:00, P009F:00 • Suction control valve: P062A:00 • Fuel pressure sensor: P0191:00, P0192:00, P0193:00 • Fuel temperature sensor: P0181:00, P0182:00, P0183:00
P009F:00	Fuel pressure relief valve malfunction (stuck on)	• When the following condition is met, the PCM detects the control current at fuel pressure relief valve as exceeding 30 A and control voltage at fuel pressure relief valve as 1.2—4.8 V for continuous 8 times. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — 0.42 msec or more have elapsed since fuel pressure relief pressure operation (fuel pressure reduction) — 2 s or more have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00

DTC No.	Condition	Detection condition
P0101:00	MAF sensor circuit range/ performance problem	• When the following conditions are met, the intake air amount is out of the specification (fluctuates with engine speed) for a continuous **5.2 s**. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **60 s** have elapsed after enigine was started. — Engine speed: **1,000—3,500 rpm** — Fuel injection amount: **0 mm^3/st** — Charge air temperature: **below 65 °C {149 °F}** — Desired HP-EGR valve position: **below 1 %** — Desired intake shutter valve position: **80 °** — Intake manifold pressure: **below 200 kPa {2.03 kgf/cm^2, 29.0 psi}** — Duty value of the suction control valve: **above 32 %** — IAT sensor No.2: Normal — BARO sensor: Normal — The following DTCs are not detected: • HP-EGR control valve position sensor: P0405:00, P0406:00 • HP-EGR control valve DC motor control system: P0404:00, P0488:00 • Intake shutter valve control: P2101:00ÅAP2118:00ÅAP1589:00 • MAF sensor: P0102:00, P0103:00 • IAT sensor No.2: P007C:00, P007D:00 • MAP sensor: P0107:00, P0108:00 • Intake sutter valve position sensor: P0122:00, P0123:00 • BARO sensor: P2228:00, P2229:00
P0102:00	MAF sensor circuit low input	• If the PCM detects that the MAF sensor voltage at the PCM terminal 2AC is **0.07 V or less** for **5 s** with the following condition met, the PCM determines that the MAF sensor circuit voltage is low. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0103:00	MAF sensor circuit high input	• The PCM monitors the input voltage from the MAF sensor. If the input voltage at the PCM terminal 2AC is **above 4.71 V** for **5 s**, the PCM determines that the MAF sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — Fuel injection amount: **200 mm^3/stroke or less**

DTC No.	Condition	Detection condition
P0106:00	MAP sensor No.2 circuit range/ performance problem	• When any of the following conditions is met: — With the following conditions met, a condition has continued for **1.8 s within 4 to 6 s** after the ignition was switched OFF in which difference between the barometric pressure and intake air pressure is **50.3 kPa {0.513 kgf/cm^2, 7.30 psi} or more** and difference between the intake air pressure and exhaust gas pressure is **50.3 kPa {0.513 kgf/cm^2, 7.30 psi} or more**. • Battery voltage: **10 V or more** • Intake shutter valve malfunction DTC P1589:00 not detected — The PCM detects a condition two times in which difference between the maximum intake air pressure when the following condition A is met and the minimum intake air pressure when the following condition B is met is l**ess than 2.1 kPa {0.021 kgf/cm^2, 0.30 psi}**. **Condition A: the following conditions continue for 0.032 s or more** • Battery voltage: **8 V or more** • **0.5 s** have elapsed after the ignition was switched ON (engine off or on) • Engine coolant temperature: **55 °C {131 °F} or more** • Intake air temperature: **-10 °C {14 °F} or more** • Barometric pressure: **65—120 kPa {0.67—1.22 kgf/cm^2, 9.5—17.4 psi}** • Change in fuel injection amount: **Less than 4 mm^3/st within 0.016 s** • Vehicle speed: **above 60 km/h {37 mph}** • Target intake air pressure is **150 kPa {1.53 kgf/cm^2, 21.8 psi} or more** for **1.5 s or more** • Difference in fuel injection amount before and after correction is **less than 5 mm^3/st** • ECT sensor malfunction DTC P0117:00, P0118:00 not detected • MAP sensor malfunction DTC P0107:00, P0108:00 not detected • VSS malfunction DTC P0500:00 not detected • Intake shutter valve position sensor malfunction DTC P0122:00, P0123:00 not detected • Intake shutter valve malfunction DTC P1589:00, P2101:00, P2118:00 not detected • Turbocharger solenoid valve malfunction DTC P0047:00, P0048:00 not detected **Condition B: the following conditions continue for 0.032 s or more** • Battery voltage: **8 V or more** • **0.5 s** have elapsed after the ignition was switched ON (engine off or on) • Engine coolant temperature: **60 °C {140 °F} or more** • Intake air temperature: **-10 °C {14 °F} or more** • Barometric pressure: **65—120 kPa {0.67—1.22 kgf/cm^2, 9.5—17.4 psi}** • Change in fuel injection amount: **Less than 4 mm^3/st within 0.016 s** • ECT sensor malfunction DTC P0117:00, P0118:00 not detected • MAP sensor malfunction DTC P0107:00, P0108:00 not detected • VSS malfunction DTC P0500:00 not detected • Intake shutter valve position sensor malfunction DTC P0122:00, P0123:00 not detected • Intake shutter valve malfunction DTC P1589:00, P2101:00, P2118:00 not detected • Turbocharger solenoid valve malfunction DTC P0047:00, P0048:00 not detected

DTC No.	Condition	Detection condition
P0107:00	MAP sensor No.2 circuit low input	• The PCM monitors the input voltage from the MAP sensor. If the input voltage at the PCM terminal 1BA is **below 0.12 V for 4 s**, the PCM determines that the MAP sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — When any of the following conditions is met: • Intake shutter valve opening angle: **above 8 °** • HP-EGR control is being performed and diesel particulate filter regeneration control is not performed
P0108:00	MAP sensor No.2 circuit high input	• The PCM monitors the input voltage from the MAP sensor. If the input voltage at the PCM terminal 1BA is **above 4.92 V** for **4 s**, the PCM determines that the MAP sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0111:00	IAT sensor No.1 circuit range/ performance problem	• With the following conditions met, a condition has continued for **15 s** in which difference in intake air temperature measured by the IAT sensor No. 1 and boost air temperature sensor is **150 °C {270 °F} or more** and difference in intake air temperature measured by IAT sensor No.1 and IAT sensor No.2 is **150 °C {270 °F} or more**. **MONITORING CONDITIONS** — Period vehicle being left: **6 h or more** — Battery voltage: **10 V or more** — **Within 20 s** after engine start — The following DTCs are not detected: • Boost air temperature sensor: P007C:00, P007D:00 • IAT sensor No.1: P0112:00, P0113:00
P0112:00	IAT sensor No.1 circuit low input	• The PCM monitors the IAT sensor No.1 signal. If the PCM detects that the IAT sensor No.1 voltage at the PCM terminal 2AS is **below 0.09 V** for **5 s**, the PCM determines that the IAT sensor No.1 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0113:00	IAT sensor No.1 circuit high input	• The PCM monitors the IAT sensor No.1 signal. If the PCM detects that the IAT sensor No.1 voltage at the PCM terminal 2AS is **above 4.61 V** for **5 s**, the PCM determines that the IAT sensor No.1 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0116:00	ECT sensor circuit range/ performance problem	• With the following conditions met, a condition has continued for **5 min** in which change in the engine coolant temperature after the fuel cut control was performed is **less than 0.004 °C {0.007 °F}**. **MONITORING CONDITIONS** — Period vehicle being left: **6 h or more** — Battery voltage: **10 V or more** — VSS: Normal — The following DTCs are not detected: • ECT sensor: P0117:00, P0118:00 • Coolant control valve position sensor: P2681:00, P2682:00, P2683:00, P26A3:00
P0117:00	ECT sensor circuit low input	• The PCM monitors the ECT sensor signal. If the PCM detects that the ECT sensor voltage at the PCM terminal 1CY is **below 0.05 V** for **5 s**, the PCM determines that the ECT sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — Ignition switched ON (engine off or on)

DTC No.	Condition	Detection condition
P0118:00	ECT sensor circuit high input	• The PCM monitors the ECT sensor signal. If the PCM detects that the ECT sensor voltage at the PCM terminal 1CY is **above 4.91 V** for **5 s**, the PCM determines that the ECT sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — Ignition switched ON (engine off or on)
P0122:00	Intake shutter valve position sensor circuit low input	• If the input voltage at the PCM terminal 1AV is **less than 0.12 V** for **2 s**, the PCM determines that the intake shutter valve position sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — Ignition switched ON (engine off or on)
P0123:00	Intake shutter valve position sensor circuit high input	• If the input voltage at the PCM terminal 1AV is **more than 4.80 V** for **2 s**, the PCM determines that the intake shutter valve position sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — Ignition switched ON (engine off or on)
P0130:00	A/F sensor circuit range/ performance problem	• With the following conditions met, a condition has continued for **5 s** in which the difference in output voltage at A/F sensor positive and negative terminals is **0.1 V or less**. **MONITORING CONDITIONS** — Output voltage at A/F sensor negative terminal: **3.8—4.7 V** — Battery voltage: **11—16 V** — The following DTCs are not detected: • A/F sensor: P0131:00, P0132:00
P0131:00	A/F sensor circuit low input	• When the following condition is met, the voltage at A/F sensor terminal A is **less than 0.73 V**, or the voltage at A/F sensor terminal B is less than 0.80 V for a continuous **5 s**. **MONITORING CONDITIONS** — Battery voltage: **11—16 V**
P0132:00	A/F sensor circuit high input	• When the following condition is met, the voltage at A/F sensor terminal A **exceeds 4.15 V** or the voltage at A/F sensor terminal B **exceeds 3.79 V** for a continuous **5 s**. **MONITORING CONDITIONS** — Battery voltage: **11—16 V**
P0134:00	A/F sensor circuit problem	• The PCM monitors the A/F sensor impedance when the following conditions are met. If the impedance is **more than 65 ohms** for **5 s**, the PCM determines that A/F sensor is not activated. **MONITORING CONDITIONS** — Battery voltage: **11—16 V** — Ignition switched ON (engine on) — A/F sensor heater feedback control: Being performed — The following DTCs are not detected: • A/F sensor: P0131:00, P0132:00
P0181:00	Fuel temperature sensor circuit range/performance problem	• With the following conditions met, a condition has continued for **3.2 s** in which the change in fuel temperature after engine start is **less than 1 1 °C {2 °F}**. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — Ignition switched ON (engine on) — Engine coolant temperature after engine start: **-25—60 °C {-13—140 °F}** — Elapsed time after engine start: **108 to 839 s** — Travel time after engine start at vehicle speed of **3 km/h {2 mph} or less**: **26 s or more** — Travel time after engine start at vehicle speed of **25 km/h {16 mph} or more**: **60 s or more** — The following DTCs are not detected: • ECT sensor: P0117:00, P0118:00 • Fuel temperature sensor: P0182:00, P0183:00 • VSS: P0500:00

DTC No.	Condition	Detection condition
P0182:00	Fuel temperature sensor circuit low input	• The PCM monitors the fuel temperature sensor signal. If the PCM detects that the fuel temperature sensor voltage at the PCM terminal 1CU is **below 0.06 V** for **1 s**, the PCM determines that the fuel temperature sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0183:00	Fuel temperature sensor circuit high input	• The PCM monitors the fuel temperature sensor signal. If the PCM detects that the fuel temperature sensor voltage at the PCM terminal 1CU is **above 4.91 V** for **1 s**, the PCM determines that the fuel temperature sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0191:00	Fuel pressure sensor circuit range/ performance problem	• The difference between the maximum and minimum value of the fuel pressure sensor output voltage in **less than 0.0025 V** or the difference between the actual fuel pressure and target fuel pressure relative to the feedback **exceeds 5 MPa {51 kgf/cm2, 725 psi}** for a continuous **1 s** when the following conditions are met. **MONITORING CONDITIONS** — Engine speed: **50—5,200 rpm or less** — Fuel temperature: **90 °C {194 °F} or less** — Target fuel pressure: **more than 20 MPa {204 kgf/cm^2, 2,901 psi}** — Target fuel injection amount: **exceeds 5 mm^3** — Target fuel discharge amount from supply pump: **less than 1,000 mm^3/rev** — The following DTCs are not detected: • Fuel pressure sensor: P0192:00, P0193:00 • Fuel temperature sensor: P0182:00, P0183:00 • Suction control valve: P062A:00 • Fuel leakage detected: P0093:00
P0192:00	Fuel pressure sensor circuit low input	• If the input voltage at the PCM terminal 1AG is **less than 0.222 V** for **0.7 s**, the PCM determines that the fuel pressure sensor circuit is low. **MONITORING CONDITIONS** — Battery voltage: **8—20 V**
P0193:00	Fuel pressure sensor circuit high input	• If the input voltage at the PCM terminal 1AG is **more than 4.492** V for **0.7 s**, the PCM determines that the fuel pressure sensor circuit is high. **MONITORING CONDITIONS** — Battery voltage: **8—20 V**
P0196:00	Engine oil temperature sensor circuit range/performance problem	• With the following conditions met, a condition has continued for **14.8 s** in which difference in temperature measured by the engine oil temperature sensor and ECT sensor is **150 °C {270 °F} or more**. **MONITORING CONDITIONS** — Period vehicle being left: **6 h or more** — Battery voltage: **10 V or more** — **Within 20 s** after engine start — The following DTCs are not detected: • ECT sensor: P0117:00, P0118:00 • Engine oil temperature sensor: P0197:00, P0198:00
P0197:00	Engine oil temperature sensor circuit low input	• The PCM monitors the input signal from the engine oil temperature sensor. If the voltage from the engine oil temperature sensor is **below 0.1 V** for **5 s**, the PCM determines that the engine oil temperature sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)

DTC No.	Condition	Detection condition
P0198:00	Engine oil temperature sensor circuit high input	• The PCM monitors the engine oil temperature sensor signal. If the PCM detects that the engine oil temperature sensor voltage at the PCM terminal 1W is **above 4.96 V** for **5 s**, the PCM determines that the engine oil temperature sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0201:00	Fuel injector No.1 circuit open	• When the following condition is met, the PCM detects the control current at fuel injector No.1 as **0.67 A or less** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — During fuel injection control — **2 s or more** have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P0202:00	Fuel injector No.2 circuit open	• When the following condition is met, the PCM detects the control current at fuel injector No.2 as **0.67 A or less** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — During fuel injection control — **2 s or more** have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P0203:00	Fuel injector No.3 circuit open	• When the following condition is met, the PCM detects the control current at fuel injector No.3 as **0.67 A or less** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — During fuel injection control — **2 s or more** have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P0204:00	Fuel injector No.4 circuit open	• When the following condition is met, the PCM detects the control current at fuel injector No.4 as **0.67 A or less** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — During fuel injection control — **2 s or more** have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P0219:00	Engine overspeed condition	• PCM detects that the engine speed is **5,670 rpm or more**.
P0234:00	Turbocharger overboost condition	• The difference between the actual intake air pressure and the target intake air pressure exceeds than the specified value for a continuous 7 s when the following conditions are met.

DTC No.	Condition	Detection condition
P026A:00	Charge aie cooler efficiency below threshold	• When the following conditions are met, the boost air temperature after passing through the charge air cooler exceeds 120 °C {248 °F} for a continuous 10 s. **MONITORING CONDITIONS** — Vehicle speed: **30 km/h {19 mph} or more** — Duty value of the electric water pump: **80 % or more** — The amount of fluctuation in the boost air temperature before and after passing through the charge air cooler for the past 10 s: **Within 50 °C {122 °F}** — The following components are normal: • IAT sensor No.1 • Battery • Electric water pump circuit • Vehicle speed signal • Intake shutter valve • HP-EGR control valve • LP-EGR control valve • Exhaust shutter valve • Exhaust gas reflux amount of EGR system • Turbocharger with variable turbine geometry • Speed sensor assy (turbocharger with variable turbine geometry) • MAP sensor
P0299:00	Turbocharger underboost condition	• The difference between the actual intake air pressure and the target intake air pressure less than the specified value for a continuous 7 s when the following conditions are met.
P02EE:00	Fuel injector No.1 circuit: shorted to each other	• When the following condition is met, the PCM detects the control current at fuel injector No.1 as **0.67 A or less** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — During fuel injection control — **2 s or more** have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P02EF:00	Fuel injector No.2 circuit: shorted to each other	• When the following condition is met, the PCM detects the control current at fuel injector No.2 as **0.67 A or less** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — During fuel injection control — **2 s or more** have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P02F0:00	Fuel injector No.3 circuit: shorted to each other	• When the following condition is met, the PCM detects the control current at fuel injector No.3 as **0.67 A or less** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — During fuel injection control — **2 s or more** have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00

DTC No.	Condition	Detection condition
P02F1:00	Fuel injector No.4 circuit: shorted to each other	• When the following condition is met, the PCM detects the control current at fuel injector No.4 as **0.67 A or less** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — During fuel injection control — **2 s or more** have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P0301:00	Cylinder No.1 misfire	• The misfire rate of specific cylinders relative to the crankshaft speed exceeds the specified value continuously for a specified period of time when all of the following conditions are met. — There is a record of driving the vehicle with all of the following conditions met. • Deceleration with accelerator pedal released from engine speed of **1,500 rpm** (from engine speed at deceleration start to **1,150 rpm**) • Fuel injection amount control: during fuel cut • Speed ratio fluctuation or time fluctuation for each cylinder is specified value or less — Fuel injection pattern or A/C relay operation condition is not switched directly before. — Post injection is not being performed during deceleration and no load — Gear is not being shifted — Change in accelerator opening angle is specified value or less — Speed ratio fluctuation for each cylinder is specified value or less — Fuel injection amount is specified value or more — Engine coolant temperature: **30 °C {86 °F} or more**
P0302:00	Cylinder No.2 misfire	
P0303:00	Cylinder No.3 misfire	
P0304:00	Cylinder No.4 misfire	
P0313:00	Misfire detected with low fuel	• “Injection amount cumulative value after lowering point E”, which is calculated by the PCM, exceeds the threshold value (larger than DTC P115A:00).
P0336:00	CKP sensor circuit range/ performance problem	• There is no CKP sensor signal input while the crankshaft rotates **20 times** with the following conditions met. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine on)
P0337:00	CKP sensor circuit problem	• CKP sensor signal is not input **56 times** while the crankshaft rotates **12 times** with the following conditions met. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine on)
P0339:00	CKP sensor circuit range/ performance problem	• When any of the following conditions is met: — The calculated value of the crank angle from the CKP sensor is compared with the crank angle while a cylinder identification is performed when the engine is restarted, and the PCM cannot detect the reverse rotation directly before the engine stalls during i-stop operation. — The PCM detects the reverse rotation pulse while the crankshaft is rotated clockwise.
P0341:00	CMP sensor circuit range/ performance problem	• There is no CMP sensor signal input while the crankshaft rotates **21 times** when the following conditions are met. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine on) — Engine speed: **700 rpm or more**
P0342:00	CMP sensor circuit problem	• CMP sensor signal is not input **5 times** while the camshaft rotates **11 times** with the following conditions met. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine on)

DTC No.	Condition	Detection condition
P0401:00	HP-EGR flow insufficient detected	• When the following condition is met, the HP-EGR volume is lower than the specification for the target value for a continuous specified time. **MONITORING CONDITIONS** — During HP-EGR control — Engine speed: **1,200—2,500 rpm** — Fuel injection amount:**4—50 mm^3/stoke** — Engine torque: **20—240 N·m** — The following DTCs are not detected: • MAP sensor: P0107:00, P0108:00 • MAF sensor: P0102:00, P0103:00 • Exhaust gas pressure sensor No.1: P0472:00, P0473:00 • Exhaust gas temperature sensor No.1: P0546:00 • Intake shutter valve control: P2101:00, P2118:00, P1589:00 • Intake shutter valve position sensor: P0122:00, P0123:00 • HP-EGR control valve DC motor control system: P0404:00, P0488:00 • HP-EGR control valve position sensor: P0405:00, P0406:00 • LP-EGR EGR control valve DC motor control system: P045B:00, P213B:00 • LP-EGR control valve position sensor: P0407:00, P0408:00 • Exhaust shutter valve DC motor control system: P0475:00, P0476:00 • Exhaust shutter valve position sensor: P048D:00, P048E:00 • Internal PCM malfunction: P062B:00 • Fuel injector: P0201:00, P0202:00, P0203:00, P0204:00, P02EE:00, P02EF:00, P02F0:00, P02F1:00, P1378:00, P1379:00, P2146:00, P2147:00, P2148:00, P2149:00, P2150:00, P2151:00
P0402:00	HP-EGR flow excessive detected	• When the following condition is met, the HP-EGR volume is higher than the specification for the target value for a continuous specified time. **MONITORING CONDITIONS** — During HP-EGR control — Engine speed: **1,200—2,500 rpm** — Fuel injection amount:**4—50 mm^3/stoke** — Engine torque: **20—240 N·m** — The following DTCs are not detected: • MAP sensor: P0107:00, P0108:00 • MAF sensor: P0102:00, P0103:00 • Exhaust gas pressure sensor No.1: P0472:00, P0473:00 • Exhaust gas temperature sensor No.1: P0546:00 • Intake shutter valve control: P2101:00, P2118:00, P1589:00 • Intake shutter valve position sensor: P0122:00, P0123:00 • HP-EGR control valve DC motor control system: P0404:00, P0488:00 • HP-EGR control valve position sensor: P0405:00, P0406:00 • LP-EGR EGR control valve DC motor control system: P045B:00, P213B:00 • LP-EGR control valve position sensor: P0407:00, P0408:00 • Exhaust shutter valve DC motor control system: P0475:00, P0476:00 • Exhaust shutter valve position sensor: P048D:00, P048E:00 • Internal PCM malfunction: P062B:00 • Fuel injector: P0201:00, P0202:00, P0203:00, P0204:00, P02EE:00, P02EF:00, P02F0:00, P02F1:00, P1378:00, P1379:00, P2146:00, P2147:00, P2148:00, P2149:00, P2150:00, P2151:00
P0404:00	Circuit malfunction in HP-EGR control valve DC motor control system	• With the following conditions met, a condition has continued for **2 s** in which the DC motor drive current of the HP-EGR control valve is more than the specified value or the temperature of the DC motor drive circuit is **175 °C {347 °F} or more**. **MONITORING CONDITIONS** — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00

DTC No.	Condition	Detection condition
P0405:00	HP-EGR control valve position sensor circuit low input	• If the input voltage at the PCM terminal 1BF is **less than 0.25 V** for **2 s**, the PCM determines that the HP-EGR control valve position sensor circuit is low. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0406:00	HP-EGR control valve position sensor circuit high input	• If the input voltage at the PCM terminal 1BF is **more than 4.84 V** for **2 s**, the PCM determines that the HP-EGR control valve position sensor circuit is high. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0407:00	LP-EGR control valve position sensor circuit low input	• If the PCM detects the LP-EGR valve position sensor voltage at the PCM terminal 1AQ is **less than 0.24 V** for **2 s** with the following condition met, the PCM determines that the LP-EGR valve position sensor circuit voltage is low. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0408:00	LP-EGR control valve position sensor circuit high input	• If the PCM detects the LP-EGR valve position sensor voltage at the PCM terminal 1AQ is **4.84 V or more** for **2 s** with the following condition met, the PCM determines that the LP-EGR valve position sensor circuit voltage is high. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0421:00	Warm up catalyst system efficiency below threshold	• The difference between the exhaust gas temperature before and after passing the catalytic converter is specified value or less when the following conditions are met: **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — Engine speed: **1,200—2,000 rpm** — Fuel injection amount: **10—25 mm^3/ stroke** — IAT (IAT sensor No.1): **–10 °C {14 °F} or more** — Barometric pressure: **65 kPa {0.67 kgf/cm^2, 9.5 psi} or more** — ECT: **58 °C {136 °F} or more** — Vehicle speed: **20—140 km/h {13.0—86.9 mph}** — Accumulated PM calculated from fuel: **0.8—4.0 g/l {0.05—0.25 lb/ft^3}** — Accumulated PM calculated from pressure applied to diesel particulate filter: **0.1—2.0 g/l {0.062—0.124 lb/ft^3}** — Traveled distance after diesel particulate filter regeneration: **36—400 km {23—248 mile}** — Exhaust gas temperature before passing catalytic converter: **160—195 °C {320—383 °F}** — Exhaust gas temperature after passing catalytic converter (at engine start): **0—200 °C {32—392 °F}** — Exhaust gas temperature after passing catalytic converter: **135—250 °C {275—482 °F}** — The following DTCs are not detected: • ECT sensor: P0117:00, P0118:00 • IAT sensor No.1: P0112:00, P0113:00 • Fuel pressure sensor: P0192:00, P0193:00 • Exhaust gas temperature sensor No.2: P2032:00, P2033:00 • Exhaust gas temperature sensor No.3: P242C:00, P242D:00 • Glow plug control system: P0670:00, P0671:00, P0672:00, P0673:00, P0674:00 • Fuel injector: P0201:00, P0202:00, P0203:00, P0204:00, P02EE:00, P02EF:00, P02F0:00, P02F1:00, P1378:00, P1379:00, P2146:00, P2147:00, P2148:00, P2149:00, P2150:00, P2151:00

DTC No.	Condition	Detection condition
P045B:00	LP-EGR control valve control: circuit malfunction	• The LP-EGR control valve drive current exceeds the specification or the LP-EGR control valve drive circuit temperature is **175 °C {347 °F} or more** for a countinuous **2 s** when the following condition is met. **MONITORING CONDITIONS** — PCM internal processor malfunction DTC P0606:00 not detected.
P0471:00	Exhaust gas pressure sensor No.1 circuit range/performance problem	• When any of the following conditions is met: — With the following conditions met, a condition has continued for **1.8 s** within **4 to 6 s** after the ignition was switched OFF in which difference between the barometric pressure and exhaust gas pressure is **50.3 kPa {0.513 kgf/cm^2, 7.30 psi} or more** and difference between the intake air pressure and exhaust gas pressure is **50.3 kPa {0.513 kgf/cm^2, 7.30 psi} or more**. • Battery voltage: **10 V or more** — The PCM detects a condition two times in which difference between the maximum exhaust gas pressure when the following condition A is met and the minimum exhaust gas pressure when the following condition B is met is less than **11 kPa {0.11 kgf/cm^2, 1.6 psi}**. **Condition A: the following conditions continue for 0.32 s or more** • Battery voltage: **8—20 V** • **0.5 s** have elapsed after the ignition was switched ON (engine on) • Engine coolant temperature: **above 55 °C {131 °F}** • Intake air temperature: **above −10 °C {14 °F}** • Barometric pressure: **65—120 kPa {0.67—1.22 kgf/cm2, 9.5—17.4 psi}** • Change in fuel injection amount: **Less than 4 mm^3/st** within **0.016 s** • Vehicle speed: **above 60 km/h {37 mph}** • Target intake air pressure is **150 kPa {1.53 kgf/cm^2, 21.8 psi} or more** for **1.5 s or more** • Difference in fuel injection amount before and after correction is **less than 5 mm3/st** • VSS malfunction DTC P0500:00 not detected • Exhaust gas pressure sensor No.1 malfunction DTC P0472:00, P0473:00 not detected • Intake shutter valve position sensor malfunction DTC P0122:00, P0123:00 not detected • Intake shutter valve malfunction DTC P1589:00, P2101:00, P2118:00 not detected • Turbocharger solenoid valve malfunction DTC P0047:00, P0048:00 not detected **Condition B: the following conditions continue for 0.32 s or more** • Battery voltage: **8—20 V** • **0.5 s** have elapsed after the ignition was switched ON (engine on) • Engine coolant temperature: **above 40 °C {104 °F}** • Intake air temperature: **above −7 °C {19 °F}** • Barometric pressure: **65—120 kPa {0.67—1.22 kgf/cm2, 9.5—17.4 psi}** • Change in fuel injection amount: **Less than 4 mm^3/st** within **0.016 s** • Fuel injection amount learning is continued for **2 s or more** • VSS malfunction DTC P0500:00 not detected • Exhaust gas pressure sensor No.1 malfunction DTC P0472:00, P0473:00 not detected • Intake shutter valve position sensor malfunction DTC P0122:00, P0123:00 not detected • Intake shutter valve malfunction DTC P1589:00, P2101:00, P2118:00 not detected • Turbocharger solenoid valve malfunction DTC P0047:00, P0048:00 not detected

DTC No.	Condition	Detection condition
P0472:00	Exhaust gas pressure sensor No.1 circuit low input	• If the input voltage at the PCM terminal 1X is **less than 0.16 V** for **5 s**, the PCM determines that the exhaust gas pressure sensor No.1 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine on)
P0473:00	Exhaust gas pressure sensor No.1 circuit high input	• If the input voltage at the PCM terminal 1X is **more than 4.81 V** for **5 s**, the PCM determines that the exhaust gas pressure sensor No.1 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine on)
P0475:00	Exhaust shutter valve circuit problem	• The exhaust shutter valve drive current exceeds the specification or the exhaust shutter valve drive circuit temperature is **175 °C {347 °F}** or more for a continuous **3 s** when the following condition is met. **MONITORING CONDITIONS** — The following DTCs are not detected: • PCM internal CPU: P0606:00
P0476:00	Exhaust shutter valve circuit range/performance problem	• When the following condition is met, the exhaust shutter valve control duty value is **90%** or more for a continuos **2s.** **MONITORING CONDITIONS** — During engine running — The following DTCs are not detected: • PCM internal CPU: P0606:00 • Exhaust shutter valve DC motor control: P0475:00 • Exhaust shutter valve position sensor: P0408D:00, P048E:00
P0480:00	Fan control module control circuit problem	• The PCM monitors the input voltage from the fan control module No.1. If the voltage at the PCM terminal 2BG remains low or high for **6 s**, the PCM determines that the fan control circuit has a malfunction.
P0488:00	Duty signal error in HP-EGR control valve control system	• When the following conditions are met, the HP-EGR control valve control duty value is **90 % or more** for a continuous **2 s**: **MONITORING CONDITIONS** — HP-EGR control valve is operating — When any of the following conditions is met: • Boost air temperature: **30 °C {86 °F} or more** • Intake air temperature: **30 °C {86 °F} or more** • Engine coolant temperature: **70 °C {158 °F} or more** — The following DTCs are not detected: • HP-EGR control valve DC motor control system: P0404:00 • Boost air temperature sensor: P007C:00, P007D:00 • IAT sensor No.1: P0112:00, P0113:00 • ECT sensor: P0117:00, P0118:00
P048D:00	Exhaust shutter valve position sensor circuit low input	• If the input voltage at the PCM terminal 1AL is **less than 0.19 V** for **5 s**, the PCM determines that the exhaust shutter valve position sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after ignition was switched on. — PCM internal processor malfunction DTC P0606:00 not detected.
P048E:00	Exhaust shutter valve position sensor circuit high input	• If the input voltage at the PCM terminal 1AL is **more than 4.77 V** for **5 s**, the PCM determines that the exhaust shutter valve position sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after ignition was switched on. — PCM internal processor malfunction DTC P0606:00 not detected.

DTC No.	Condition	Detection condition
P049B:00	LP-EGR flow insufficient detected	• When the following condition is met, the LP-EGR volume is lower than the specification for the target value for a continuous specified time. **MONITORING CONDITIONS** — During LP-EGR control — During HP-EGR control — Fuel injection amount:**15—50 mm^3/stoke** — Engine speed: **1,300—2,500 rpm** — Engine torque: **70—240 N·m** — The following DTCs are not detected: • MAP sensor: P0107:00, P0108:00 • MAF sensor: P0102:00, P0103:00 • Exhaust gas pressure sensor No.1: P0472:00, P0473:00 • Exhaust gas temperature sensor No.1: P0546:00 • Intake shutter valve control: P2101:00, P2118:00, P1589:00 • Intake shutter valve position sensor: P0122:00, P0123:00 • HP-EGR control valve DC motor control system: P0404:00, P0488:00 • HP-EGR control valve position sensor: P0405:00, P0406:00 • LP-EGR control valve DC motor control system: P045B:00, P213B:00 • LP-EGR control valve position sensor: P0407:00, P0408:00 • Exhaust shutter valve DC motor control system: P0475:00, P0476:00 • Exhaust shutter valve position sensor: P048D:00, P048E:00 • Internal PCM malfunction: P062B:00 • Fuel injector: P0201:00, P0202:00, P0203:00, P0204:00, P02EE:00, P02EF:00, P02F0:00, P02F1:00, P1378:00, P1379:00, P2146:00, P2147:00, P2148:00, P2149:00, P2150:00, P2151:00
P049C:00	LP-EGR flow excessive detected	• When the following condition is met, the LP-EGR volume is higher than the specification for the target value for a continuous specified time. **MONITORING CONDITIONS** — During LP-EGR control — During HP-EGR control — Fuel injection amount:**15—50 mm^3/stoke** — Engine speed: **1,300—2,500 rpm** — Engine torque: **70—240 N·m** — The following DTCs are not detected: • MAP sensor: P0107:00, P0108:00 • MAF sensor: P0102:00, P0103:00 • Exhaust gas pressure sensor No.1: P0472:00, P0473:00 • Exhaust gas temperature sensor No.1: P0546:00 • Intake shutter valve control: P2101:00, P2118:00, P1589:00 • Intake shutter valve position sensor: P0122:00, P0123:00 • HP-EGR control valve DC motor control system: P0404:00, P0488:00 • HP-EGR control valve position sensor: P0405:00, P0406:00 • LP-EGR control valve DC motor control system: P045B:00, P213B:00 • LP-EGR control valve position sensor: P0407:00, P0408:00 • Exhaust shutter valve DC motor control system: P0475:00, P0476:00 • Exhaust shutter valve position sensor: P048D:00, P048E:00 • Internal PCM malfunction: P062B:00 • Fuel injector: P0201:00, P0202:00, P0203:00, P0204:00, P02EE:00, P02EF:00, P02F0:00, P02F1:00, P1378:00, P1379:00, P2146:00, P2147:00, P2148:00, P2149:00, P2150:00, P2151:00
P0500:00	VSS circuit problem	• With the following conditions met, a condition has continued for **5 s** in which an error signal from the DSC HU/CM is received or no signal can be received from the DSC HU/CM. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off) or the engine was started.
P0504:00	Brake switch circuit problem	• The condition in which the brake switch No.1 and No.2 signals are both on or off for **3 s or more** is repeated five times when performing the brake operation.

DTC No.	Condition	Detection condition
P0522:00	Engine oil pressure sensor circuit low input	• If the input voltage at the PCM terminal 1AB is **less than 0.14 V** for **2.5 s**, the PCM determines that the engine oil pressure sensor circuit is low. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine on)
P0523:00	Engine oil pressure sensor circuit high input	• If the input voltage at the PCM terminal 1AB is **more than 4.81 V** for **2.5 s**, the PCM determines that the engine oil temperature sensor circuit is high. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine on)
P0524:00	Engine oil pressure too low	• If the engine oil pressure is **less 30 kPa {0.31 kgf/cm^2, 4.4 psi}** for a continuous **3 s** since the engine was started.
P0532:00	Refrigerant pressure sensor circuit low input	• The PCM monitors the input voltage from the refrigerant pressure sensor. If the input voltage at the PCM terminal 2BH is **below 0.06 V** for **5 s**, the PCM determines that the refrigerant pressure sensor circuit has a malfunction.
P0533:00	Refrigerant pressure sensor circuit high input	• The PCM monitors the input voltage from the refrigerant pressure sensor. If the input voltage at the PCM terminal 2BH is **above 4.95 V** for **5 s**, the PCM determines that the refrigerant pressure sensor circuit has a malfunction.
P053B:00	Blow-by heater relay control circuit low input	• With the following conditions met, the blow-by heater power supply voltage is continuously lower than the specification for **5 s**. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — Blow-by heater relay controlled on
P053C:00	Blow-by heater relay control circuit high input	• The blow-by heater circuit current **exceeds 5.9 A** for a continuous **5 s** when the following conditions are met. **MONITORING CONDITIONS** — Battery voltage: **8—16 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — Blow-by heater relay controlled off
P0545:00	Exhaust gas temperature sensor No.1 circuit low input	• The PCM monitors the exhaust gas temperature sensor No.1 signal. If the PCM detects that the exhaust gas temperature sensor No.1 voltage at the PCM terminal 1CI is **below 0.14 V** for **5 s**, the PCM determines that the exhaust gas temperature sensor No.1 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine on)
P0546:00	Exhaust gas temperature sensor No.1 circuit high input	• The PCM monitors the exhaust gas temperature sensor No.1 signal. If the PCM detects that the exhaust gas temperature sensor No.1 voltage at the PCM terminal 1CI is **above 4.96 V** for **10 s**, the PCM determines that the exhaust gas temperature sensor No.1 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — Between the elapsed time of **14 to 70 min.** after the ignition is switched on (engine on). — Engine speed: **above 700 rpm** — Engine coolant temperature: **above 60 °C {140 °F}** — Boost air temperature: **above 5 °C {41 °F}** — Intake air temperature: **above 5 °C {41 °F}** — Vehicle speed: **above 25 km/h {16 mph}** continues for **10 min or more** — Fuel injection amount: **16 mm^3/stroke or more** — Total fuel injection amount: **500,000 mm^3 or more** — The following DTCs are not detected: • Boost air temperature sensor: P007C:00, P007D:00 • IAT sensor No.1: P0112:00, P0113:00 • ECT sensor: P0117:00, P0118:00 • VSS: P0500:00

DTC No.	Condition	Detection condition
P0555:00	Power brake unit vacuum sensor circuit problem	• The PCM monitors the power brake unit vacuum sensor signal voltage. If the PCM detects the power brake unit vacuum sensor voltage is **below 0.077 V** or **above 4.96 V**, the PCM determines that the power brake unit vacuum sensor circuit has problem.
P055F:00	Engine oil pressure control circuit low oil pressure	• The engine oil pressure is the specified value or less after the specified time has elapsed since the engine was started. — Engine speed is **2,300 rpm or less**: **70 kPa {0.71 kgf/cm2, 10 psi}** — Engine speed is **2,300 to 2,500 rpm**: **70—90 kPa {0.72—0.91 kgf/cm^2, 11—13 psi}** (engine speed is proportional to specified value of engine oil pressure) — Engine speed is **2,500 to 3,500 rpm**: **90—172.5 kPa {0.92—1.75 kgf/cm^2, 14.00—25.01 psi}** (engine speed is proportional to specified value of engine oil pressure) — Engine speed is **3,500 to 5,500 rpm**: **172.5—182.5 kPa {1.759—1.860 kgf/cm^2, 25.02—26.46 psi}** (engine speed is proportional to specified value of engine oil pressure)
P057F:00	Power system: Battery deterioration	• The battery charge/discharge flow exceeds the specification.
P058A:00	Current sensor: Function malfunction	• Error signal from the current sensor is received.
P0601:00	PCM memory check sum error	• PCM internal memory check sum error for **5 s**.
P0602:00	PCM programming error	• The vehicle model information is not recorded in the PCM for a continuous **1 s** when the following condition is met. **MONITORING CONDITIONS** — Ignition switched ON (engine off or on)
P0605:00	PCM memory check sum error	• With the following conditions met, a PCM internal memory check sum error is detected three times in a series. **MONITORING CONDITIONS** — Battery voltage: **8 V or more**
P0606:00	PCM processor error	• With the following conditions met, the PCM internal main processor has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8 V or more**
P0607:00	Control module performance problem	• With the following conditions met, the PCM internal sub processor has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8 V or more**
P0610:00	PCM vehicle configuration error	• The vehicle model information is not recorded correctly in the PCM for a continuous **1 s** when the following condition is met. **MONITORING CONDITIONS** — Ignition switched ON (engine off or on)
P0615:00	Starter relay circuit problem	• The PCM records DTCs when the number of times the engine has been started reaches the warrantied performance frequency for the starter or starter relay.
P062A:00	Suction control valve circuit problem	• Any of following conditions occurs: — With the following conditions met, a condition has continued for **1 s** in which the suction control valve control current is **1.0 A or less** or the control voltage is more than the specified value. **MONITORING CONDITIONS** • Battery voltage: **8 V or more** • **0.5 s** have elapsed after the ignition was switched ON (engine on) — With the following conditions met, a condition has continued for **1 s** in which the suction control valve control voltage is the specified value or less or the control current is **more than 1.0 A**. **MONITORING CONDITIONS** • Battery voltage: **8 V or more** • **0.5 s** have elapsed after the ignition was switched ON (engine on)

DTC No.	Condition	Detection condition
P062B:00	PCM internal malfunction	• When the following conditions are met, an operation malfunction of the clock circuit in the PCM or a communication error of the fuel injector control internal processor is detected for **1 s** or more (engine stopped) or **8 times** consecutively (engine running). **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **2 s or more** have elapsed since PCM launch — PCM internal processor malfunction DTC P0606:00 not detected.
P0642:00	Constant voltage power supply circuit low input	• When the following condition is met, the output voltage of the 5 V power supply terminal **4.63 V or less** for a continuous **1 s**: **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0643:00	Constant voltage power supply circuit high input	• When the following condition is met, the output voltage of the 5 V power supply terminal **exceeds 5.38 V** for a continuous **1 s**: **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0646:00	A/C relay circuit low input	• When the following conditions are met, PCM detects that the A/C relay voltage is **approx. 0 V** for a continuous **5 s**: **MONITORING CONDITIONS** — Battery voltage: **9—16 V** — A/C relay controlled off
P0647:00	A/C relay circuit high input	• The current value flowing to the PCM 2AQ terminal is **1 A or more** for a continuous **5 s** when any of the following conditions is met. **MONITORING CONDITIONS** — Battery voltage: **9—16 V** — A/C relay controlled on
P0652:00	Constant voltage power supply control circuit low input	• When the following condition is met, the output voltage of the 5 V power supply terminal is **4.78 V or less** for a continuous **1 s**: **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0653:00	Constant voltage power supply control circuit high input	• When the following condition is met, the output voltage of the 5 V power supply terminal **exceeds 5.23 V** for a continuous **1 s**: **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P0670:00	Glow control module control circuit problem	• When the following conditions are met, a condition continues for **5 s** in which the power supply voltage of the glow control module is **less than 5.8 V** or **exceeds 16.5 V**, or there is an internal circuit malfunction, or the temperature of the internal circuit exceeds the specification. **MONITORING CONDITIONS** — 0.5 s have elapsed after ignition was switched ON. (engine off or on) — During glow control — Battery voltage: **8—20 V** — Glow plug No.1 contol circuit voltage: **3—11.3 V** — The following DTC is not detected: • LIN communication system: U0106:00

DTC No.	Condition	Detection condition
P0671:00	Glow plug No.1 control circuit problem	• When the following conditions are met, a condition continues for **5 s** in which the current at the glow plug No.1 circuit is **less than 2.88 A** while glow control is being executed at a control voltage of **75% or more** of the battery voltage, or the at the glow plug No.1 circuit **exceeds 65 A** while glow control is being executed at a control voltage of **less than 75%** of the battery voltage. **MONITORING CONDITIONS** — 0.5 s have elapsed after ignition was switched ON. (engine off or on) — During glow control — Glow plug No.1 contol circuit voltage: **3—11.3 V** — Battery voltage: **8—20 V** — The following DTC is not detected: • LIN communication system: U0106:00
P0672:00	Glow plug No.2 control circuit problem	• When the following conditions are met, a condition continues for **5 s** in which the current at the glow plug No.2 circuit is **less than 2.88 A** while glow control is being executed at a control voltage of **75% or more** of the battery voltage, or the at the glow plug No.2 circuit **exceeds 65 A** while glow control is being executed at a control voltage of **less than 75%** of the battery voltage. **MONITORING CONDITIONS** — 0.5 s have elapsed after ignition was switched ON. (engine off or on) — During glow control — Glow plug No.1 contol circuit voltage: **3—11.3 V** — Battery voltage: **8—20 V** — The following DTC is not detected: • LIN communication system: U0106:00
P0673:00	Glow plug No.3 control circuit problem	• When the following conditions are met, a condition continues for **5 s** in which the current at the glow plug No.3 circuit is **less than 2.88 A** while glow control is being executed at a control voltage of **75% or more** of the battery voltage, or the at the glow plug No.3 circuit **exceeds 65 A** while glow control is being executed at a control voltage of **less than 75%** of the battery voltage. **MONITORING CONDITIONS** — 0.5 s have elapsed after ignition was switched ON. (engine off or on) — During glow control — Glow plug No.1 contol circuit voltage: **3—11.3 V** — Battery voltage: **8—20 V** — The following DTC is not detected: • LIN communication system: U0106:00
P0674:00	Glow plug No.4 control circuit problem	• When the following conditions are met, a condition continues for **5 s** in which the current at the glow plug No.4 circuit is **less than 2.88 A** while glow control is being executed at a control voltage of **75% or more** of the battery voltage, or the at the glow plug No.4 circuit **exceeds 65 A** while glow control is being executed at a control voltage of **less than 75%** of the battery voltage. **MONITORING CONDITIONS** — 0.5 s have elapsed after ignition was switched ON. (engine off or on) — During glow control — Glow plug No.1 contol circuit voltage: **3—11.3 V** — Battery voltage: **8—20 V** — The following DTC is not detected: • LIN communication system: U0106:00
P0698:00	Constant voltage power supply circuit low input	• When the following condition is met, the output voltage of the **5 V** power supply terminal **2.45 V** or less for a continuous **1 s**: **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (under the engine on or off)
P0699:00	Constant voltage power supply circuit high input	• When the following condition is met, the output voltage of the **5 V** power supply terminal exceeds **5.56 V** for a continuous **1 s**: **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (under the engine on or off)

DTC No.	Condition	Detection condition
P06B8:00	Internal control module non-volatile RAM error	• PCM internal EEPROM malfunction.
P06DB:00	Engine oil solenoid valve circuit low input	• If the PCM detects that the engine oil solenoid valve control voltage at the PCM terminal 1CO is specified value or **less for 5 s** with the following condition met, the PCM determines that the engine oil solenoid valve circuit voltage is low. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — Engine oil solenoid valve control duty value: **94% or less**
P06DC:00	Engine oil solenoid valve control circuit high input	• If the PCM detects that the engine oil solenoid valve control current at the PCM terminal 1CO is **3.5 A or more** for **5 s** with the following condition met, the PCM determines that the engine oil solenoid valve circuit voltage is high. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — Engine oil solenoid valve control duty value: **5% or more**
P06DD:00	Engine oil pressure switch control circuit high input	• When the following condition is met, the engine oil pressure is **exceeds 200 kPa {2.04 kgf/cm2, 29.0 psi}** for **5 s**. **MONITORING CONDITIONS** — Engine oil temperature sensor: Normal — Engine oil pressure sensor: Normal — During low hydraulic pressure control (during engine oil solenoid valve operation) — Engine oil temperature: **20 °C {68 °F} or more** — Engine speed is specified value or more. • Idle speed or more when engine oil temperature is **20 °C {68 °F} or less** (when cold) • **2,000 rpm or more** when engine oil temperature is **90 °C {194 °F}** (when hot) • **4,000 rpm or more** when engine oil temperature is **135 °C {275 °F} or more** (when hot)
P06DE:00	Engine oil pressure switch control circuit low input	• When the following condition is met, the engine oil pressure is **200 kPa {2.04 kgf/cm2, 29.0 psi} or less** for **5 s**. **MONITORING CONDITIONS** — Engine oil temperature sensor: Normal — Engine oil pressure sensor: Normal — During high hydraulic pressure control (during engine oil solenoid valve operation) — Engine oil temperature: **20 °C {68 °F} or more** — Engine speed is specified value or more. • Idle speed or more when engine oil temperature is **20 °C {68 °F} or less** (when cold) • **2,000 rpm or more** when engine oil temperature is **90 °C {194 °F}** (when hot) • **4,000 rpm or more** when engine oil temperature is **135 °C {275 °F} or more** (when hot) — The following DTCs are not detected: • Malfunction due to no engine oil pressure: P0524:00 • Malfunction due to engine oil pressure decrease: P055F:00
P0703:00	Brake switch input circuit problem	• The PCM monitors the input signal from the brake switch No.1. If the input signal does not change while following decelerating **8 times**, the PCM determines that there is a brake switch No.1 input circuit problem. **MONITORING CONDITIONS** — Vehicle speed: from **above 30 km/h {19 mph}** to **0 km/h {0 mph}** — Deceleration rate: **exceeds 2.35 km/h {1.46 mph} per sec**

DTC No.	Condition	Detection condition
P0704:00*1	CPP switch input circuit problem	• The PCM monitors changes in input voltage from the CPP switch. If the PCM does not detect a voltage change while the vehicle runs with vehicle speed **above 30 km/h {19 mph}** and stops (**3 s or more**) **8 times** alternately, the PCM determines that the CPP switch circuit has a malfunction. **MONITORING CONDITIONS** — The following DTCs are not detected: • VSS: P0500:00
P07BE:00*1	Transmission indeterminate failure (failed to neutral)	• Correlation malfunction between neutral switches No.1 and No.2.
P0850:00*1	Neutral switch No.1 input circuit problem	• The PCM monitors changes in input voltage from the neutral switch No.1. If the PCM does not detect a voltage change while driving the vehicle at a vehicle speed **above 30 km/h {19 mph}** and clutch pedal is pressed and released **10 times** repeatedly, the PCM determines that the neutral switch No.1 circuit has a malfunction. **MONITORING CONDITIONS** — The following DTCs are not detected: • CPP switch: P0704:00 • VSS: P0500:00
P0A06:00	Electric water pump control circuit low input	• When the following conditions are met, the accumulated time under a condition in which the electric water pump control circuit voltage is **less than 35%** of the battery voltage is **5 s or more.** **MONITORING CONDITIONS** — All of the following conditions is met for 1min: — Duty value of the electric water pump: **80 % or more** • Battery voltage: **11—20 V** • Duty value of the electric water pump: **30—100 %**
P0A07:00	Electric water pump control circuit low input	• When the following conditions are met, the accumulated time under a condition in which the electric water pump control current exceeds **5.9 A is 5 s or more**. **MONITORING CONDITIONS** — All of the following conditions is met for 1min: • Battery voltage: **11—16 V** • Duty value of the electric water pump: **0—85 %**
P0A0F:00	Engine failed to restart	• Any of following conditions occurs: — When the engine should restart after it is stopped by the i-stop control, it does not start even though it is cranked for **3 s or more**. — When the engine should restart after it is stopped by the i-stop control, it does not cranked.
P0A8D:00	Power supply system circuit low input	**Without i-ELOOP** • Any one of the following conditions is met: — Battery voltage is **7.25 V or less** when engine restarts from i-stop. — DC-DC converter terminal voltage is **6 V or less** when engine restarts from i-stop. — PCM terminal voltage is **6.2 V or less** when engine starts. — Battery internal resistance is specified value or more. — Battery voltage is **11 V or less** when engine is switched off. **With i-ELOOP** • Any one of the following conditions is met: — Battery voltage is **7.25 V or less** when engine restarts from i-stop. — PCM terminal voltage is **6.2 V or less** when engine starts. — Battery internal resistance is specified value or more. — Battery voltage is **11 V or less** after **30 s** since system began switch to relay mode
P0A8F:00	Power system: Low input	• Any one of the following conditions is met: — Battery voltage is **11 V or less** when ignition is switched off after leaving vehicle for **30 days or more**. — Battery voltage is **11 V or less** for a continuous **5 days** when ignition is switched off.

DTC No.	Condition	Detection condition
P0A94:00	**Without i-ELOOP** • DC-DC converter: control circuit signal error **With i-ELOOP** • DC-DC converter (i-ELOOP): control circuit signal error	**Without i-ELOOP** • Internal malfunction signal from DC-DC converter via body control module (BCM) is received.(CAN/LIN communication). • Input signal from the DC-DC converter limits the pressure increase time. • Input signal from the DC-DC converter does not implement pressure increase after a pressure increase command to the DC-DC converter. **With i-ELOOP** • Internal malfunction signal from DC-DC converter (i-ELOOP) is received. • Condition in which DC-DC converter (i-ELOOP) output voltage, based on demand voltage of PCM, is low continues. • Condition in which the PCM receives a relay operation command signal from the DC-DC converter (i-ELOOP) which differs from the relay operation signal sent from the DC-DC converter (i-ELOOP). Or the signal itself is not received. • Condition in which DC-DC converter (i-ELOOP) 4A terminal current value of **30 A or more** continues.
P0C73:00	Electric water pump control: abnormal operation	• When the following monitoring conditions are met, the PCM terminal 1CP voltage is approx. **6.0 V or more** for a continuous **1 s**: **MONITORING CONDITIONS** — Engine speed: Is the engine speed in the span of the **3 to 6 s** following the **3 s** from when **700 rpm** or more is reached. — Electric water pump control duty value: **10—20%** — Battery voltage: **8—16V** — The following DTCs are not detected: • IAT sensor No.1: P0112:00, P0113:00 • Electric water pump circuit: P0A06:00, P0A07:00
P1051:00	Injection quantity too low cylinder No.1	• Small torque fluctuation condition continues in No.1 cylinder while main injection is performed **40 times** in cylinder No.1 by fuel injector No.1
P1052:00	Injection quantity too high cylinder No.1	• Large torque fluctuation condition continues in No.1 cylinder while main injection is performed **40 times** in cylinder No.1 by fuel injector No.1
P1053:00	Injection quantity too low cylinder No.2	• Small torque fluctuation condition continues in No.2 cylinder while main injection is performed **40 times** in cylinder No.2 by fuel injector No.2
P1054:00	Injection quantity too high cylinder No.2	• Large torque fluctuation condition continues in No.2 cylinder while main injection is performed **40 times** in cylinder No.2 by fuel injector No.2
P1055:00	Injection quantity too low cylinder No.3	• Small torque fluctuation condition continues in No.3 cylinder while main injection is performed **40 times** in cylinder No.3 by fuel injector No.3
P1056:00	Injection quantity too high cylinder No.3	• Large torque fluctuation condition continues in No.3 cylinder while main injection is performed **40 times** in cylinder No.3 by fuel injector No.3
P1057:00	Injection quantity too low cylinder No.4	• Small torque fluctuation condition continues in No.4 cylinder while main injection is performed **40 times** in cylinder No.4 by fuel injector No.4
P1058:00	Injection quantity too high cylinder No.4	• Large torque fluctuation condition continues in No.4 cylinder while main injection is performed **40 times** in cylinder No.4 by fuel injector No.4
P111A:00	Engine coolant temperature is high	• The engine coolant temperature is **110 °C {230 °F} or more** for **2.5 s**.
P1142:00	Fuel Restriction Condition	• Any one of the following conditions is met: — Value of exhaust gas pressure sensor No. 4 is specified value or more*1 — Turbocharger with variable turbine geometry operated in excessive speed — Turbocharger with variable turbine geometry overboost condition*1 *1: The conditions vary according to the engine speed and the operation condition of the turbocharger with variable turbine geometry.
P115A:00	Lack of remaining fuel (output restriction)	• “Injection amount cumulative value after lowering point E”, which is calculated by the PCM, exceeds the threshold value.
P115B:00	Lack of remaining fuel (fuel injection pause)	• “Injection amount cumulative value after lowering point E”, which is calculated by the PCM, exceeds the threshold value (larger than DTC P0313:00).
P117A:00	Record of torque restriction for piston protection	• The engine oil pressure is the specified value or less for a continuous **15 s**. • The vehicle is driven continuously for a certain period of time at low engine speed and high load.
P1200:00	Fuel injection amount learning not completed	• Fuel injector injection amount correction is not completed correctly.
P1260:00	Immobilizer system problem	• The start stop unit detects an immobilizer system malfunction.

DTC No.	Condition	Detection condition
P1282:00	Common rail control system problem	• When the following conditions are met, the common rail fuel pressure value **exceeds 250 MPa {2,549 kgf/cm^2, 36,260 psi}** or the fuel pressure value exceeds the value calculated by the PCM based on the engine speed for a continuous **1 s**. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine on) — Engine speed: **above 500 rpm** — The following DTCs are not detected: • Fuel pressure sensor: P0191:00, P0192:00, P0193:00
P1284:00	Fuel injector carbon deposit burn-off procedure not completed	• The fuel injector carbon deposit burn-off procedure was not completed normally.
P1329:00	Common rail pressure higher than desired (engine running)	• When the following conditions are met, the common rail fuel pressure value **exceeds 250 MPa {2,549 kgf/cm^2, 36,260 psi}** or the fuel pressure value exceeds the value calculated by the PCM based on the engine speed for a continuous **1 s**. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine on) — The following DTCs are not detected: • Fuel pressure sensor: P0191:00, P0192:00, P0193:00 • When the following conditions are met, the common rail fuel pressure value exceeds **217 MPa {2,213 kgf/cm^2, 31,473 psi}** or the fuel pressure value exceeds the value calculated by the PCM based on the engine speed for a continuous **60 s**. **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **0.5 s** have elapsed after the ignition was switched ON (engine on) — The following DTCs are not detected: • Fuel pressure sensor: P0191:00, P0192:00, P0193:00
P1336:00	CKP sensor/CMP sensor malfunction	• If the phase difference between the CKP sensor and CMP sensor exceeds the specified value, the PCM determines that the timing chain is stretched and stores DTC P1336:00.
P1378:00	Fuel injector circuit low input	• When the following conditions are met, a condition in which the charge voltage for the fuel indicator control is **less than 38.6 V** continues for **0.1 s**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — **2 s or more** have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P1379:00	Fuel injector circuit high input	• When the following conditions are met, a condition in which the charge voltage for the fuel indicator control **exceeds 61.5 V** continues (engine off) for **32.6 s**, or it is detected 255 times consecutively (engine running) **MONITORING CONDITIONS** — Battery voltage: **8 V or more** — **2 s or more** have elapsed since PCM launch — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00

DTC No.	Condition	Detection condition
P1589:00	Intake shutter valve control duty signal error	• The actual intake shutter valve opening angle is **4.2° larger** than the target opening angle for a continuous **5 s** when the following conditions are met. — The following conditions continue for **2 s**: • Battery voltage: **8 V or more** • Target intake shutter valve opening angle: **below 75 °** • Target intake shutter valve opening angle is in the closed direction. • Change in speed of the target intake shutter valve: **Less than 0.25 °/msec.** — Intake shutter valve control: Being performed • The actual intake shutter valve opening angle is **4.2 ° smaller** than the target opening angle for a continuous **5 s** when the following conditions are met. — The following conditions continue for **2 s**: • Battery voltage: **8 V or more** • Target intake shutter valve opening angle: **below 75 °** • Target intake shutter valve opening angle is in the closed direction. • Change in speed of the target intake shutter valve: **Less than 0.25 °/msec.** — Intake shutter valve control: Being performed
P1675:00	Fuel injector code not programmed	• The fuel injector identification number is not programmed into the PCM for a continuous **1 s** when the following condition is met. **MONITORING CONDITIONS** — Ignition switched ON (engine off or on)
P1676:00	Fuel injector code mistakenly programmed	• The fuel injector identification number is mistakenly programmed into the PCM for a continuous **1 s** when the following condition is met. **MONITORING CONDITIONS** — Ignition switched ON (engine off or on)
P167B:00	Fuel injection amount learning not completed in fuel injection system	• Fuel injector injection amount correction is not completed.
P176E:00*1	Clutch stroke sensor/starter interlock switch correlation problem	• If the clutch stroke sensor circuit voltage **above 4.86 V** or **below 0.14 V** for **5 s**, the PCM determines that the clutch stroke sensor circuit has a malfunction. • Correlation error between the clutch stroke sensor and the CPP switch, starter interlock switch is detected when any of the following conditions is met. — Correlation error content: • CPP switch is ON and clutch pedal stroke amount is **less than 1%** • CPP switch is OFF and clutch pedal stroke amount is **larger than 70%** • Starter interlock switch is ON and clutch pedal stroke amount is **less than 10%** • CPP switch is OFF and starter interlock switch is ON **MONITORING CONDITIONS** — The following DTCs are not detected: • CPP switch: P0704:00
P1905:00	Check connector circuit low input	• PCM detects that the check connector voltage is **approx. 0 V** for a continuous **0.016 s**.

DTC No.	Condition	Detection condition
P2002:00	Diesel particulate filter function decreased	• The difference in the pressure before and after passing the diesel particulate filter of less than the specified value is detected for **2s**. **MONITORING CONDITIONS** — All of the following conditions are met for **2s**: • APP: **0.5% or less** • Fuel injection amount control: during fuel cut • Amount of exhaust gas: **1.1 m^3/min or more** • Selector lever drive position and clutch engagement (MTX) • Selector lever drive range (ATX) • 300 s or more elapsed since DPF regeneration control stopped • Engine coolant temperature after engine start: **40 °C {104 °F} or more** • Intake air temperature: **–10—110 °C {14—230 °F}** • Barometric pressure: **70—110 kPa {0.72—1.12 kgf/cm^2, 11.0—15.9 psi}** • Amount of change in engine speed: **40 rpm or less** for **1 s** • Amount of change in exhaust gas: **1.8 m^3/min or more** for **1 s** • Amount of change in pressure applied to diesel particulate filter: **0.62 kPa {0.0063 kgf/cm^2, 0.090 psi} or less** for **1 s** • The traveled distance after DPF regeneration exceeds **21 km {14 mile}** and **10 s** have elapsed with DPF temperature of **350 °C {662 °F} or more**, or traveled distance **exceeds 500 km {311 mile}**. • Traveled distance after DPF regeneration **exceeds 1000 km {621.4 mile}** or accumulated PM amount **exceeds 3.4 g/l {0.22 lb/ft^3}**, or it is **less than 1 g/l {0.062 lb/ft^3}**
P2032:00	Exhaust gas temperature sensor No.2 circuit low input	• The PCM monitors the exhaust gas temperature sensor No.2 signal. If the PCM detects that the exhaust gas temperature sensor No.2 voltage at the PCM terminal 1CA is **below 0.07 V** for **5 s**, the PCM determines that the exhaust gas temperature sensor No.2 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P2033:00	Exhaust gas temperature sensor No.2 circuit high input	• The PCM monitors the exhaust gas temperature sensor No.2 signal. If the PCM detects that the exhaust gas temperature sensor No.2 voltage at the PCM terminal 1CA is **above 4.96 V** for **10 s**, the PCM determines that the exhaust gas temperature sensor No.2 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — Between the elapsed time of **14 to 70 min.** after the ignition is switched on (engine on). — Engine speed: **above 700 rpm** — Engine coolant temperature: **above 60 °C {140 °F}** — Charge air temperature: above **-10 °C {14 °F}** — Intake air temperature: above **-10 °C {14 °F}** — Vehicle speed: **above 25 km/h {16 mph}** continues for **10 min or more** — Fuel injection amount: **16 mm^3/st or more** — Total fuel injection amount: **250,000 mm^3 or more** — IAT sensor No.2: Normal — The following DTCs are not detected: • IAT sensor No.1: P0112:00, P0113:00 • ECT sensor: P0117:00, P0118:00 • IAT sensor No.2: P007C:00, P007D:00 • VSS: P0500:00

DTC No.	Condition	Detection condition
P2101:00	Intake shutter valve control circuit overcurrent	• When any of the following conditions is met: — The intake shutter valve drive current exceeds the specification for a continuous **2 s** when the following condition is met. **MONITORING CONDITIONS** • PCM internal processor malfunction DTC P0606:00 not detected — With the following conditions met, a condition has continued for **2 s** in which the temperature of the DC motor drive circuit is **175 °C {347 °F} or more**. **MONITORING CONDITIONS** • PCM internal processor malfunction DTC P0606:00 not detected
P2104:00	Intake shutter valve control system: Idle air control	• When all of the following conditions are met, the PCM determines that the exhaust gas pressure or boost pressure is the estimated value or more. — Ignition switched ON (engine on). — Shift position: except P or N position (ATX) — Shift lever position is neutral (MTX)
P2105:00	Intake air shutter valve actuator control system malfunction (forced engine shutdown)	• After switching the ignition off, the PCM detects that the engine speed has not decreased for a specified time.
P2106:00	Throttle Actuator Control System - Forced Limited Power	• Exhaust gas pressure or boost pressure of specified value or more is detected.
P2118:00	Intake shutter valve control duty signal error	• The intake air shutter valve control duty value is **90 %** for a continuous **2 s**. **MONITORING CONDITIONS** — The following DTCs are not detected: • Intake shutter valve position sensor: P0122:00, P0123:00 • Intake shutter valve: P1589:00, P2101:00
P2122:00	APP sensor No.1 circuit low input	• The PCM monitors the input voltage from APP sensor No.1. If the input voltage at the PCM terminal 2AN is **below 0.29 V** for **0.5 s**, the PCM determines that the APP sensor No.1 circuit input voltage is low. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P2123:00	APP sensor No.1 circuit high input	• The PCM monitors the input voltage from APP sensor No.1. If the input voltage at the PCM terminal 2AN is **above 4.83 V** for **0.5 s**, the PCM determines that the APP sensor No.1 circuit input voltage is high. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P2127:00	APP sensor No.2 circuit low input	• The PCM monitors the input voltage from APP sensor No.2. If the input voltage at the PCM terminal 2AX is **less than 0.22 V** for **0.5 s**, the PCM determines that the APP sensor No.2 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P2128:00	APP sensor No.2 circuit high input	• The PCM monitors the input voltage from APP sensor No.2. If the input voltage at the PCM terminal 2AX is **more than 4.15 V** for **0.5 s**, the PCM determines that the APP sensor No.2 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P2138:00	APP sensor No.1/No.2 correlation problem	• With the following conditions met, the output voltage of APP sensors No.1 and No.2 is compared, and the difference in the voltage is **0.7 V or more** for a continuous **1 s**. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)

DTC No.	Condition	Detection condition
P213B:00	LP-EGR valve control: high duty malfunction	• When the following conditions are met, if the LP-EGR control valve duty value is **90 % or more** for **2 s**, the PCM determines that there is a LP-EGR control valve control system range/performance problem. **MONITORING CONDITIONS** — LP-EGR control: during feedback control — The following DTCs are not detected: • LP-EGR EGR control valve DC motor control system: P045B:00 • LP-EGR control valve position sensor: P0407:00, P0408:00
P2146:00	Fuel injector No.1 and No.4 circuit abnormal operation	• When the following condition is met, the PCM detects the control current at fuel injectors No.1 and No.4 as **less than the 0.67 A** value for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — **2 s or more** have elapsed since PCM launch — During fuel injection control — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P2147:00	Fuel injector No.1 and No.4 circuit low input	• When the following condition is met, the PCM detects the control voltage at fuel injectors No.1 and No.4 as **1.2 V or less** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — **2 s or more** have elapsed since PCM launch — During fuel injection control — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P2148:00	Fuel injector No.1 and No.4 circuit high input	• When the following condition is met, the PCM detects the control voltage at fuel injectors No.1 and No.4 as 4.8 V or more for continuous 8 times. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — **2 s or more** have elapsed since PCM launch — During fuel injection control — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P2149:00	Fuel injector No.2 and No.3 circuit abnormal operation	• When the following condition is met, the PCM detects the control current at fuel injectors No.2 and No.3 as **less than the 0.67 A** value for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — **2 s or more** have elapsed since PCM launch — During fuel injection control — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P2150:00	Fuel injector No.2 and No.3 circuit low input	• When the following condition is met, the PCM detects the control voltage at fuel injectors No.2 and No.3 as **1.2 V or less** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — **2 s or more** have elapsed since PCM launch — During fuel injection control — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00

DTC No.	Condition	Detection condition
P2151:00	Fuel injector No.2 and No.3 circuit high input	• When the following condition is met, the PCM detects the control voltage at fuel injectors No.2 and No.3 as **4.8 V or more** for continuous **8 times**. **MONITORING CONDITIONS** — During engine running — Battery voltage: **8 V or more** — **2 s or more** have elapsed since PCM launch — During fuel injection control — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • Internal PCM malfunction: P062B:00
P2227:00	BARO sensor circuit range/ performance problem	• With the following conditions met, a condition has continued for **1.8 s within 7 to 9 s** after the ignition was switched OFF in which difference between the barometric pressure and intake air pressure is **50.3 kPa {0.513 kgf/ cm^2, 7.30 psi} or more** and difference between the barometric pressure and exhaust gas pressure is **50.3 kPa {0.513 kgf/cm^2, 7.30 psi} or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — The following DTCs are not detected: • MAP sensor: P0106:00 • Intake shutter valve: P1589:00 • BARO sensor: P2228:00, P2229:00
P2228:00	BARO sensor circuit low input	• The PCM monitors input voltage from the BARO sensor. If the input voltage is **below 1.17 V** for **5 s**, the PCM determines that the BARO sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P2229:00	BARO sensor circuit high input	• The PCM monitors input voltage from the BARO sensor. If the input voltage is **above 4.66 V** for **5 s**, the PCM determines that the BARO sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P2263:00	Regulating valve control circuit range/performance problem	• The following conditions are met, the difference between the target regulating valve opening angle and the actual regulating valve opening angle is **2 mm {0.08 in}** or more for a continuous 3 s: — Engine speed: above **700 rpm** — Turbocharger solenoid valve control duty value: **55 %** or more — Actuator position sensor: normal — HP-EGR control valve: Fully closed — Diesel particulate filter regeneration control is not performed • The following conditions are met, the difference between the target regulating valve opening angle and the actual regulating valve opening angle is **-6.2 mm {-0.25 in}** or less for a continuous 3 s: — Engine speed: above **700 rpm** — Turbocharger solenoid valve control duty value: **20 %** or less — Actuator position sensor: normal — HP-EGR control valve: Fully closed — Diesel particulate filter regeneration control is not performed
P2299:00	Accelerator pedal: spring back malfunction	• When under the following conditions it is detected that the brake pedal is depressed during driving for the specified time. — Racing — Engine speed: **1,150 rpm or more**

DTC No.	Condition	Detection condition
P242C:00	Exhaust gas temperature sensor No.3 circuit low input	• The PCM monitors the exhaust gas temperature sensor No.3 signal. If the PCM detects that the exhaust gas temperature sensor No.3 voltage at the PCM terminal 1BW is **below 0.07 V** for **5 s**, the PCM determines that the exhaust gas temperature sensor No.3 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P242D:00	Exhaust gas temperature sensor No.3 circuit high input	• The PCM monitors the exhaust gas temperature sensor No.3 signal. If the PCM detects that the exhaust gas temperature sensor No.3 voltage at the PCM terminal 1BW is **above 4.96 V** for **10 s**, the PCM determines that the exhaust gas temperature sensor No.3 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — Between the elapsed time of 1**4 to 70 min.** after the ignition is switched on (engine on). — Engine speed: **above 700 rpm** — Engine coolant temperature: **above 60 °C {140 °F}** — Intake air temperature: **-10 °C {14 °F}** — Charge air temperature: **-10 °C {14 °F}** — Vehicle speed: **above 25 km/h {16 mph}** continues for **10 min or more** — Fuel injection amount: **16 mm^3/st or more** — Total fuel injection amount: **250,000 mm^3** or more — The following DTCs are not detected: • IAT sensor No.1: P0112:00, P0113:00 • ECT sensor: P0117:00, P0118:00 • IAT sensor No.2: P007C:00, P007D:00 • VSS: P0500:00
P242F:00	Diesel particulate filter excess accumulation (exceeded acceptable amount)	• Amount of accumulated soot **exceeds 17 g/l {1.06 lb/ft^3}** because of diesel particulate filter clogging.
P243C:00	Abnormal diesel particulate filter regeneration interval short distance	• The following conditions are met **3 times**: — Drive distance from previous diesel particulate filter regeneration to next diesel particulate filter regeneration is **50 km {31 mile} or less** — PM target accumulation amount is high relative to actual PM regeneration amount
P244A:00	Exhaust gas pressure sensor No.2 range/performance problem	• With the following conditions met, a condition has continued for **7 s** in which the difference in the pressure before and after passing the diesel particulate filter is less than the specified value. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — The following conditions continue for 3 s: • Engine speed: **825 rpm or more** • Amount of exhaust gas: **2.5 m^3/min or more** • Change in exhaust gas amount: **0.03 m^3/min or less** within **0.016 s** • Exhaust gas temperature detected by exhaust gas temperature sensors No.2 and No.3: **500 °C {932 °F} or less** • Exhaust gas temperature estimated by PCM based on signal from exhaust gas temperature sensor No.4: **500 °C {932 °F} or less** • PM accumulation amount: **1.5 g/l {0.094 lb/ft^3} or more**
P2452:00	Signal malfunction in exhaust gas pressure sensor No.2	• Difference in pressure is **±5 kPa {±0.05 kgf/cm^2, ±0.7 psi} or more** while the engine is stopped.
P2454:00	Exhaust gas pressure sensor No.2 circuit low input	• If the input voltage at the PCM terminal 1AC is **less than 0.07 V** for **5 s**, the PCM determines that the exhaust gas pressure sensor No.2 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)

DTC No.	Condition	Detection condition
P2455:00	Exhaust gas pressure sensor No.2 circuit high input	• If the input voltage at the PCM terminal 1AC is **more than 4.93 V** for **5 s,** the PCM determines that the exhaust gas pressure sensor No.2 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P2456:00	Characteristic malfunction in exhaust gas pressure sensor No.2	• With the following conditions met, a condition has continued while the crankshaft rotates eight times in which the difference in diesel particulate filter pressure between that of when the exhaust gas amount is **2.0 m^3/min or more** and that of when the exhaust gas amount is less than **0.85 m^3/min** is **less than 0.1 kPa {0.001 kgf/cm^2, 0.01 psi}**. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on) — The following DTCs are not detected: • MAF sensor: P0102:00, P0103:00 • Exhaust gas pressure sensor No.2: P2454:00, P2455:00
P2458:00	Diesel particulate filter excess accumulation (mid)	• PCM detects that the soot accumulation amount **above 10 g/l {0.6 lb/ft^3}**.
P2459:00	Abnormal diesel particulate filter regeneration interval short distance*3	• After driving the vehicle at a PM accumulation amount equivalent to **3 g/l {0.187 lb/ft^3} or more**, the distance the vehicle was driven from the previous diesel particulate filter regeneration to the next diesel particulate filter regeneration is any of the following. — **39—69 km {25—42 mile}** occurs **44 times or more** — **70—99 km {44—61 mile}** occurs **115 times or more**
P2463:00	Diesel particulate filter excess accumulation (large)	• PCM detects that the soot accumulation amount **above 13 g/l {0.8 lb/ft^3}**.
P246C:00	Engine oil dilution, oil pressure decreased	• The amount of engine oil dilution exceeds the specified value (fluctuates by ambient temperature).
P2470:00	Exhaust gas temperature sensor No.4 circuit low input	• If the PCM detects that the exhaust gas temperature sensor No.4 voltage at the PCM terminal 1CE is **0.07 V or less** for **5 s** with the following condition met, the PCM determines that the exhaust gas temperature sensor No.4 circuit voltage is low. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P2471:00	Exhaust gas temperature sensor No.4 circuit high input	• The PCM monitors the exhaust gas temperature sensor No.4 signal. If the PCM detects that the exhaust gas temperature sensor No.4 voltage at the PCM terminal 1CE is **above 4.96 V** for **10 s**, the PCM determines that the exhaust gas temperature sensor No.4 circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — Between the elapsed time of **14 to 70 min.** after the ignition is switched on (engine on). — Engine speed: **above 700 rpm** — Fuel injection amount: **16 mm^3/st or more** — Total fuel injection amount: **250,000 mm^3** or more — Engine coolant temperature: **above 60 °C {140 °F}** — Intake air temperature: **-10 °C {14 °F}** — Charge air temperature: **-10 °C {14 °F}** — Vehicle speed: **above 25 km/h {16 mph}** continues for **10 min or more** — The following DTCs are not detected: • IAT sensor No.1: P0112:00, P0113:00 • ECT sensor: P0117:00, P0118:00 • IAT sensor No.2: P007C:00, P007D:00 • VSS: P0500:00

DTC No.	Condition	Detection condition
P2502:00	Generator system: Malfunction in voltage generated by generator	**Without i-ELOOP** • The voltage generated by the generator is **17 V or higher** and the battery voltage is **11 V or less** for a continuous specified time. **With i-ELOOP** • Battery voltage is **11 V or less** and condition in which current sensor detects battery charging for specified period continues
P2503:00	Generator system: Voltage generated by generator is low	**Without i-ELOOP** • A condition continues for a specified period of time in which the target generator output current calculated by the PCM is **20 A or more** and the generator output voltage is **8.5 V or less**. **With i-ELOOP** • Any one of the following conditions is met: — Field current demand output to generator is normal range and condition in which field current returned from generator is in non-power generation range continues for specified period — Condition in which malfunction status signal from generator is in returned condition continues for specified period
P2504:00	Generator system: Voltage generated by generator is high	**Without i-ELOOP** • The voltage generated by the generator is **18.5 V or higher** or the battery voltage is **16 V or higher** for a continuous specified time. **With i-ELOOP** • Any one of the following conditions is met: — Field current value returned by generator is abnormally larger than field current value demanded of generator, or condition in which battery voltage is **15.9 V or more** continues for specified period — Capacitor (i-ELOOP) voltage is **26.8 V or more**, or condition in which over-voltage notification signal from generator continues for specified period. — Condition in which overheating detection signal output from generator continues for specified period.
P2507:00	PCM battery voltage low input	• When the following conditions are met, a condition in which the backup power voltage is **less than 50%** of the battery voltage continues for **10 s**.. **MONITORING CONDITIONS** — Battery voltage: **10—16 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P250A:00	Engine oil level signal: engine oil level sensor malfunction	• PCM receives error signal from engine oil level sensor.
P250C:00	Engine oil level sensor power supply circuit malfunction	• When the following conditions are met, a supply voltage error signal from the engine oil level sensor continues to be received for a specified time. **MONITORING CONDITIONS** — All of the following conditions is met for 1min. • Battery voltage: **10—16 V** • Communication error between oil level sensor and PCM DTC U1100:00 not detected.
P250F:00	Engine oil level signal: engine oil level low	• The oil level has decreased below TBD. (TBD mm below L mark on dipstick)
P252F:00	Engine oil level too high	• The amount of engine oil dilution **exceeds 1,661 g {58.59 oz}** during the diesel particulate filter regeneration control.

DTC No.	Condition	Detection condition
P253F:00	Engine oil deteriorated	• If any of the following conditions are met: — The amount of engine oil dilution exceeds the specified value during the diesel particulate filter regeneration control. (European (L.H.D./ U.K.)) — The amount of engine oil dilution exceeds the specified value (fluctuates by ambient temperature). — With the following driving conditions met, the engine oil pressure decreases **51 kPa {0.52 kgf/cm^2, 7.4 psi} or more** compared to when the engine oil was replaced: **MONITORING CONDITIONS** • Engine speed: **1,275—1,325 rpm** • Engine oil temperature: **89.5—90.5 °C {193.1—194.9 °F}** • Traveled distance after engine oil replacement is specified value* or more *: Specified value: • **12,000 km {7,456.5 mile} or more** (European (L.H.D.)/Israel) • **7,000 km {4,350 mile} or more** (except for European (L.H.D.)/Israel)
P2564:00	Regulating valve position sensor circuit low input	• If the input voltage at the PCM terminal 1BK is **less than 0.10 V** for **1 s**, the PCM determines that the actuator position sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P2565:00	Regulating valve position sensor circuit high input	• If the input voltage at the PCM terminal 1BK is **more than 4.64 V** for **1 s**, the PCM determines that the actuator position sensor circuit has a malfunction. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — **0.5 s** have elapsed after the ignition was switched ON (engine off or on)
P2579:00	Speed sensor assy. (turbocharger) range/performance problem	• When the following conditions are met, the PCM detects **5 times** that the turbine speed detected by the speed sensor component (turbocharger with variable turbine geometry) is **lower than 17,000 rpm or more** compared to the turbine speed estimated by the PCM. **MONITORING CONDITIONS** — Gear position: 3rd or 5th — Vehicle speed: **30—110 km/h {19.0—68.3 mph}** — Intake air temperature: **20—35 °C {68—95 °F}** — APP: **30—50 %** — Engine coolant temperature: **80 °C {176 °F} or more** — Barometric pressure: **98 kPa {736 mmHg, 29 inHg} or more** — DPF regeneration control is not performed
P2580:00	Speed sensor assy. (turbocharger) circuit low input	• If the input voltage at the PCM terminal 1EJ is **less than 2.5 V** for **3.5 s**, the PCM determines that the speed sensor assy. (turbocharger) circuit is low. **MONITORING CONDITIONS** — **0.5 s** have elapsed after the engine was started.
P2581:00	Speed sensor assy. (turbocharger) circuit high input	• If the input voltage at the PCM terminal 1EJ is **more than 2.5 V** for **3.5 s**, the PCM determines that the speed sensor assy. (turbocharger) circuit is high. **MONITORING CONDITIONS** — **0.5 s** have elapsed after the engine was started.
P2610:00	Instrument cluster internal engine off timer performance problem	• The PCM detected an instrument cluster internal malfunction for the specified period of time when any of the following conditions is met.. **MONITORING CONDITIONS** — The following DTCs are not detected: • CKP sensor: P0336:00, P0337:00 • Camshaft position/Crankshaft position correlation problem: P0016:00
P2681:00	Circuit malfunction in coolant control valve control system	• The coolant control valve drive current exceeds the specification or drive circuit temperature is **175 °C {347 °F} or more** for a continuous **3 s** when the following condition is met: **MONITORING CONDITIONS** — PCM internal processor malfunction DTC P0606:00 not detected.

DTC No.	Condition	Detection condition
P2682:00	Coolant control valve position sensor circuit low input	• The PCM monitors the coolant control valve position sensor signal. When the following conditions are met, if the PCM detects that the coolant control valve position sensor voltage at the PCM terminal 1BP is below **0.66 V** for **5 s**, the PCM determines that the coolant control valve position sensor circuit is low. **MONITORING CONDITIONS** — **0.5 s** have elapsed after the ignition was switched ON (under the engine on or off)
P2683:00	Coolant control valve position sensor circuit high input	• The PCM monitors the coolant control valve position sensor signal. When the following conditions are met, if the input voltage at the PCM terminal 1BP is above **4.89 V** for **5 s**, the PCM determines that the coolant control valve position sensor circuit input voltage is high. **MONITORING CONDITIONS** — **0.5 s** have elapsed after the ignition was switched ON (under the engine on or off)
P26A3:00	Coolant control valve control: high duty malfunction	• When the following conditions are met, the engine coolant control valve actuator control duty value determined by the engine coolant temperature and intake air temperature exceeds the threshold for a continuous specified time. **MONITORING CONDITIONS** — Engine coolant temperature: **-30—120 °C {-22—248 °F}** — Battery voltage: **8 V or more** — During coolant control valve control — The following DTCs are not detected: • PCM internal processor malfunction: P0606:00 • ECT sensor: P0117:00, P0118:00 • Coolant control valve position sensor: P2682:00, P2683:00
P2A00:00	Atmosphere learning malfunction in A/F sensor circuit	• With the following conditions met, a condition has continued for **5 s** in which the A/F sensor output value is deviated from the preset standard value by **36.5% or more** **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — Engine speed: 800 rpm or more — A/F sensor feedback correction is actuated
U0073:00	CAN system communication error (HS CAN)	• With the following conditions met, a CAN bus communication line error is continued for **1 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on)
U0074:00*2	CAN system communication error (local CAN between PCM and TCM)	• With the following conditions met, a CAN bus communication line error is continued for **1 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on)
U0101:00*2	CAN communication: communication error to TCM	• With the following conditions met, a communication error between the PCM and TCM is continued for **5 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on)

DTC No.	Condition	Detection condition
U0106:00	LIN communication system: glow plug control module information communication error with PCM	• With the following conditions continued for **1.5 s**, a communication error between the glow control module is continued for **5 s** or more. **MONITORING CONDITIONS** — Battery voltage: above **8 V** — Ignition switch is ON (under the engine on or off) • With the following conditions met, a communication error between the glow control module is continued for 5 s or more. **MONITORING CONDITIONS** — Battery voltage: **8—20 V** — 0.5 s have elapsed after the ignition was switched ON (under the engine on or off) — Voltage at glow plug-related circuit is **6 to 11 V** continuously for **0.2 s or more**
U0121:00	CAN communication: communication error to DSC HU/CM	• With the following conditions met, a communication error between the PCM and DSC HU/CM is continued for **5 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on)
U0131:00	CAN communication: communication error to EPS control module	• With the following conditions met, a communication error between the PCM and EPS control module is continued for **5 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on)
U0140:00	CAN communication: communication error to front body control module (FBCM)	• With the following conditions met, a communication error between the PCM and BCM is continued for **5 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on)
U0151:00	CAN communication: communication error to SAS control module	• With the following conditions met, a communication error between the PCM and SAS control module is continued for **5 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on)
U0155:00	CAN communication: communication error to instrument cluster	• With the following conditions met, a communication error between the PCM and instrument cluster is continued for **5 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on)
U0214:00	CAN communication: communication error to start stop unit	• With the following conditions met, a communication error between the PCM and start stop unit is continued for **5 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on)
U0235:00	CAN communication: communication error to laser sensor	• With the following conditions met, a communication error between the PCM and laser sensor is continued for **5 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on)

DTC No.	Condition	Detection condition
U0298:00	**Without i-ELOOP** • CAN/LIN communication system: DC-DC converter information communication error with BCM **With i-ELOOP** • CAN communication: communication error to DC-DC converter (i-ELOOP)	**Without i-ELOOP** • With the following conditions met, a communication error between the PCM and DC-DC converter is continued for **0.75 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on) **With i-ELOOP** • With the following conditions met, a communication error between the PCM and DC-DC converter (i-ELOOP) is continued for **5 s or more**. **MONITORING CONDITIONS** — Battery voltage: **10 V or more** — **1.5 s** have elapsed after the ignition was switched ON (engine off or on)
U0302:00*2	TCM processor error	• With the battery voltage **10 V or more**, any of the following conditions is continuously met for **3.5 s**. — Check sum error in signal from TCM — Counter error in signal from TCM
U0315:00	DSC HU/CM error	• With the battery voltage **10 V or more**, any of the following conditions is continuously met for **3.5 s**. — Check sum error in signal from DSC HU/CM — Counter error in signal from DSC HU/CM
U0320:00	EPS control module error	• With the battery voltage **10 V or more**, any of the following conditions is continuously met for **3.5 s**. — Check sum error in signal from EPS control module — Counter error in signal from EPS control module
U0323:00	Instrument cluster error	• With the battery voltage **10 V or more**, any of the following conditions is continuously met for **3.5 s**. — Counter error in signal from instrument cluster
U0336:00	SAS control module error	• With the battery voltage **10 V or more**, any of the following conditions is continuously met for **3.5 s**. — Check sum error in signal from SAS control module — Counter error in signal from SAS control module
U0338:00	Start stop unit error	• If any of the following conditions are met: — With the battery voltage **10 V or more**, any of the following conditions is continuously met for **3.5 s**. • Check sum error in signal from start stop unit • Counter error in signal from start stop unit — PCM receives error signal from start stop unit for **1 s**.
U0433:00	Abnormal message from laser sensor	• Correct data cannot be received from laser sensor.
U0599:00	Invalid data received from DC-DC converter	• Invalid data is received from DC-DC converter.
U1007:00	CAN/LIN communication system: current sensor information communication error with body control module (BCM)	• When the following conditions are met, the PCM detects a current sensor data communication error from the BCM for **0.8 s**. — Battery voltage: **10 V or more** — Condition continues for **3 s** in which battery voltage decreases to **less than 15 V** from condition **exceeding 15 V**
U1100:00	LIN communication: communication error to engine oil level sensor	• With the following conditions continued for **1.5 s**, a communication error between the PCM and engine oil level sensor is continued for **5 s** or more. — Battery voltage: **10—16 V** — Ignition switched ON (engine off or on)
U2300:00	Global central configuration error	• With the battery voltage more than specified value, the following conditions is continuously met for **5 s**. — No configuration of the instrument cluster. — The configuration signal with the estimated CAN ID is not sent from the instrument cluster. — The configuration signal value sent via CAN from the instrument cluster is unknown or invalid. — The configuration signal value sent via CAN from the instrument cluster is a value other than the estimated value. — The configuration signal value sent via CAN from the instrument cluster does not match the PCM value.

DTC No.	Condition	Detection condition
U3000:41	PCM processor error	• The following conditions are met: — Switch the ignition ON (engine off or on). — Data related to the immobilizer system fails to write to the non-volatile memory in the PCM. — The code related to the immobilizer system stored in the non-volatile memory cannot be read normally.

*1 : MTX
*2 : ATX
*3 : Glow indicator light flashes